

AVIRAL GARG

Lexington, KY | (859) 433-9706 | aviralk.garg@gmail.com | [linkedin.com/in/aviral-garg-011273208](https://www.linkedin.com/in/aviral-garg-011273208)

SUMMARY

Computer Science graduate (M.S., University of Kentucky) with hands-on experience spanning machine learning, full-stack development, and cloud-native distributed systems. Builds and ships ML/GenAI pipelines, RAG applications, and production software — from reinforcement-learning and deep-learning models in PyTorch/TensorFlow to React front ends, Kubernetes deployments, and SQL/NoSQL data layers. Brings a strong foundation in algorithms and distributed systems, backed by prior industry experience on clinical diagnostic hardware.

EDUCATION

M.S., Computer Science — University of Kentucky, Lexington, KY Jan 2024 – May 2026

B.Tech., Electronics & Computers Engineering — Thapar Institute of Engineering & Technology, India May 2018 – Jun 2022

TECHNICAL SKILLS

Languages: Python, C++, C#, C, JavaScript, SQL, R, Solidity, HTML5/CSS3, Bash

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Transformers, Hugging Face, Gradio, Deep Reinforcement Learning (DQN, PPO), NLP (NLTK, spaCy), RAG, LLMs, Statistical Modeling

Cloud & Data Engineering: AWS, Docker, Kubernetes, Knative, Serverless Architecture, Distributed Systems (RPC), NumPy, Pandas, Grafana, Network Security, Linux

Web, Full-Stack & Databases: React, React Native, Flask, .NET, Android Studio, Firebase, Google Maps API, PostgreSQL, MySQL, MongoDB, DBMS Integration

PROFESSIONAL EXPERIENCE

Student Engineering Technician — University of Kentucky Feb 2026 – May 2026

- Engineered Python-based data-automation and visualization pipelines with a custom graphical user interface (GUI), standardizing quality assurance and quality control (QA/QC) workflows for high-precision laboratory research.
- Calibrated, maintained, and troubleshooted mechanical/electrical assemblies and high-throughput data-acquisition hardware while leading a multidisciplinary team of undergraduate engineers, whose work supported a peer-reviewed ASABE publication.

Front End Developer — Dept. of Computer Science, University of Kentucky Sep 2024 – Jan 2025

- Architected intuitive UI/UX workflows for Generative AI applications using LlamaIndex and FAISS vector databases to enable efficient Retrieval-Augmented Generation (RAG) over large document sets.
- Integrated backend database management system (DBMS) architecture and built NLP tokenization and text-cleaning pipelines to preprocess unstructured data, eliminating indexing redundancies.

Software Engineer, R&D — Recorders & Medicare Systems, Panchkula, India Jul 2022 – Jun 2023

- Designed and optimized .NET-based desktop interfaces for high-precision medical diagnostic equipment, including EEG, ECG, and X-ray systems used in clinical settings.
- Leveraged SQL and MongoDB for efficient data storage and retrieval in embedded medical systems, and managed a secure virtualized environment to isolate, test, and deploy critical software releases.

PROJECTS

Risk-Aware AI Portfolio Management System (GenAI/RL) Jan 2026 – May 2026

- Designed an autonomous reinforcement-learning trading agent (Deep Q-Networks) with configurable risk limits and sector-diversification constraints across multiple asset classes.
- Fused quantitative market data with qualitative news signals using Transformer-based models and a retrieval-augmented generation (RAG) pipeline for automated risk auditing.

Dynamic Portfolio Management with Bandit Algorithms (ML) Aug 2025 – Dec 2025

- Implemented LinUCB and Proximal Policy Optimization (PPO) frameworks to trade multi-asset portfolios, benchmarking strategies in Python with NumPy/Pandas on Sharpe ratio and maximum drawdown.

Knative Function-Chain Autoscaling & Stateful File Server (Cloud/DevOps) Aug 2025 – Dec 2025

- Evaluated Knative autoscaling policies for chained, containerized functions on Kubernetes/Docker, and built a Sun RPC-based stateful file server with XDR consistency guarantees.

Fake News Detection on Social Media (NLP) Jan 2025 – May 2025

- Built a supervised classification pipeline (BERT, SVM, Naive Bayes) to classify the credibility of social-media content, using custom BeautifulSoup/Scrapy web-scraping workflows and NLTK/spaCy preprocessing.